

Lösungen – Terme und Gleichungen (Fit für Klasse 10)

Rechenwege verkürzt dargestellt. Ergebnisse **fett**. Bei F2 steht D für die Diskriminante $b^2 - 4ac$.

A · Terme zusammenfassen

A1

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|--------------------|----------------|----------------|
| a) $11x$ | b) $7a$ | c) $3x + 7y$ |
| d) $6a + 5b$ | e) $2x^2 + 7x$ | f) $9ab + 3a$ |
| g) $9x + 7$ | h) $3x^2 + 2$ | i) $13x - 10y$ |
| j) $4a^2 + 5a + 3$ | k) $3ab + 10b$ | l) $2x^2$ |

A2

- | | | |
|------------|-----------|------------|
| a) $12x^2$ | b) $10ab$ | c) $-6x^2$ |
| d) x^3 | e) $6x^3$ | f) $8a^2$ |

A3 ($x = 3, y = -2$)

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|-------------------|----------------|---------------------------------|
| a) $12 - 4 = 8$ | b) $9 - 4 = 5$ | c) $3 \cdot 3 \cdot (-2) = -18$ |
| d) $18 + 10 = 28$ | | |

B · Klammern auflösen

B1

- | | | |
|---|--|---|
| a) $3x + 12$ | b) $10a - 15$ | c) $-2x + 12$ |
| d) $12x + 8 - 5x = 7x + 8$ | e) $2a + 6 + 3a - 3 = 5a + 3$ | f) $6x - 2x + 7 = 4x + 7$ |
| g) $8x + 4 - 3x + 6 = 5x + 10$ | h) $6a - 12 - 4a + 12 = 2a$ | i) $x^2 + 5x + 2x + 10 = x^2 + 7x + 10$ |
| j) $2x^2 + 8x - 3x - 12 = 2x^2 + 5x - 12$ | k) $x^2 + 7x + 3x + 21 = x^2 + 10x + 21$ | |
| l) $3x^2 - 15x + 2x - 10 = 3x^2 - 13x - 10$ | | |

B2

- | | | |
|------------------|------------------|---------------------|
| a) $6x^2 + 8x$ | b) $-6a^2 + 15a$ | c) $x^3 - 4x^2 + x$ |
| d) $8ab + 12b^2$ | | |

C · Binomische Formeln

C1

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|-----------------------------|---------------------------------|----------------------------|
| a) $x^2 + 6x + 9$ (1. bF) | b) $x^2 - 10x + 25$ (2. bF) | c) $4x^2 + 4x + 1$ (1. bF) |
| d) $x^2 - 16$ (3. bF) | e) $9a^2 - 12ab + 4b^2$ (2. bF) | f) $25 - x^2$ (3. bF) |
| g) $x^2 + 12x + 36$ (1. bF) | h) $16x^2 - 24x + 9$ (2. bF) | |

C2

- | | | |
|-----------------------|-----------------------|-----------------|
| a) $(x + 4)^2$ | b) $(x + 7)(x - 7)$ | c) $(2x - 3)^2$ |
| d) $(a - 3)^2$ | e) $(x + 10)(x - 10)$ | f) $(3x + 1)^2$ |
| g) $(5a + 4)(5a - 4)$ | h) $(x + 1)^2$ | |

C3

- | | | |
|---------------------------------|---------------------------------|--------------------------------|
| a) $x^2 + 10x + 25 = (x + 5)^2$ | b) $x^2 - 12x + 36 = (x - 6)^2$ | c) $x^2 - 81 = (x + 9)(x - 9)$ |
| d) $x^2 + 14x + 49 = (x + 7)^2$ | | |

D · Faktorisieren (Ausklammern)

D1

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|-----------------|-------------------|------------------|
| a) $7(x + 3)$ | b) $4(3a - 2)$ | c) $5x(x + 2)$ |
| d) $3b(2a - 3)$ | e) $x(x + 5)$ | f) $4xy(2x + 3)$ |
| g) $5x(3x - 1)$ | h) $2ab(2a + 3b)$ | |

D2

- a) $3(x^2 - 9) = 3(x + 3)(x - 3)$ (3. bF)
 b) $2(x^2 + 4x + 4) = 2(x + 2)^2$ (1. bF)
 c) $5(x^2 - 4) = 5(x + 2)(x - 2)$ (3. bF)
 d) $3(x^2 - 6x + 9) = 3(x - 3)^2$ (2. bF)

D3

a) $\frac{3(2x+3)}{3} = 2x + 3$

b) $\frac{x(x+5)}{x} = x + 5$

c) $\frac{4a(a-3)}{4a} = a - 3$

E · Lineare Gleichungen**E1**

- a) $| - 7 \rightarrow x = 8$ b) $| : 4 \rightarrow x = 9$ Probe: $4 \cdot 9 = 36 \checkmark$ c) $| - 5, | : 3 \rightarrow x = -7$
 d) $3x = 15 \rightarrow x = 5$ e) $4x = 24 \rightarrow x = 6$
 f) $5x - 10 = 3x + 8 \rightarrow 2x = 18 \rightarrow x = 9$ Probe: $5 \cdot 7 = 35 = 27 + 8 \checkmark$ g) $5x - 4 = 21 \rightarrow 5x = 25 \rightarrow x = 5$
 h) $\frac{x}{4} = 4 \rightarrow x = 16$ i) $8x + 4 = 5x + 19 \rightarrow 3x = 15 \rightarrow x = 5$
 j) $6x - 8 = 4x + 4 \rightarrow 2x = 12 \rightarrow x = 6$ k) $4x = -16 \rightarrow x = -4$
 l) $6x - 15 = 4x + 1 \rightarrow 2x = 16 \rightarrow x = 8$ m) $\frac{x}{3} = 6 \rightarrow x = 18$
 n) $3x + 8 = 23 \rightarrow 3x = 15 \rightarrow x = 5$

E2

- a) $3x - 7 = 23 \rightarrow 3x = 30 \rightarrow x = 10$
 b) $x + 12 = 4x \rightarrow 12 = 3x \rightarrow x = 4$
 c) Breite b , Länge $2b$: $2 \cdot (b + 2b) = 36 \rightarrow 6b = 36 \rightarrow b = 6 \text{ cm}$, Länge = 12 cm

F · Quadratische Gleichungen**F1**

- a) $x_1 = 9; x_2 = -9$ b) $x^2 = 49 \rightarrow x_1 = 7; x_2 = -7$ c) $x^2 = 25 \rightarrow x_1 = 5; x_2 = -5$
 d) $x^2 = 16 \rightarrow x_1 = 4; x_2 = -4$ e) $x^2 = \frac{1}{4} \rightarrow x_1 = 0,5; x_2 = -0,5$
 f) $x^2 = -9 \rightarrow$ **keine Lösung** (Quadrat nie negativ) g) $x^2 = 121 \rightarrow x_1 = 11; x_2 = -11$
 h) $x^2 = 9 \rightarrow x_1 = 3; x_2 = -3$

F2

- a) $a = 1, b = 5, c = 6$; $D = 25 - 24 = 1 \rightarrow x_{1,2} = \frac{-5 \pm 1}{2} \rightarrow x_1 = -2; x_2 = -3$
 b) $a = 1, b = -2, c = -15$; $D = 4 + 60 = 64 \rightarrow x_{1,2} = \frac{2 \pm 8}{2} \rightarrow x_1 = 5; x_2 = -3$
 c) $a = 2, b = 7, c = 3$; $D = 49 - 24 = 25 \rightarrow x_{1,2} = \frac{-7 \pm 5}{4} \rightarrow x_1 = -0,5; x_2 = -3$
 d) $a = 1, b = -6, c = 9$; $D = 36 - 36 = 0 \rightarrow x = 3$ (genau eine Lösung, da $D = 0$)
 e) $a = 1, b = 4, c = 5$; $D = 16 - 20 = -4 < 0 \rightarrow$ **keine Lösung** (negative Diskriminante)
 f) $a = 1, b = 1, c = -12$; $D = 1 + 48 = 49 \rightarrow x_{1,2} = \frac{-1 \pm 7}{2} \rightarrow x_1 = 3; x_2 = -4$
 g) $a = 2, b = -8, c = 6$; $D = 64 - 48 = 16 \rightarrow x_{1,2} = \frac{8 \pm 4}{4} \rightarrow x_1 = 3; x_2 = 1$
 h) $a = 1, b = 10, c = 25$; $D = 100 - 100 = 0 \rightarrow x = -5$ (genau eine Lösung)

F3

- a) $D = 9 + 28 = 37 > 0 \rightarrow$ **zwei Lösungen**
 b) $D = 64 - 64 = 0 \rightarrow$ **genau eine Lösung**
 c) $D = 1 - 32 = -31 < 0 \rightarrow$ **keine Lösung**